WeResence.

TECHNICAL SPEC • TRD v1.0.0

WeResence — Technical Requirement Document

Cryptographic Specifications, Port 8901 Dispatching & Zero-Knowledge Architecture

> **Architectural Mandate:** End-to-end zero-knowledge security across heterogeneous transport layers. All messages are encrypted locally using the Double Ratchet protocol over Curve25519 before hitting any transport socket or radio transmitter. The server and telecom carriers remain strictly dumb bit-pipes.

1. Cryptographic Protocol Specification

Layer	Algorithm	Key Lifecycle
Key Agreement	X3DH (Extended Triple Diffie-Hellman)	Curve25519 Ephemeral + Signed Pre-keys
Forward Secrecy	Double Ratchet (Signal protocol)	KDF chain step on every single message
Symmetric Cipher	AES-256-GCM / ChaCha20-Poly1305	Unique 96-bit IV + 128-bit authentication tag
Local Storage	SQLCipher (AES-256 in CBC/GCM mode)	PBKDF2 with 64,000 iterations

2. Port 8901 Binary Telephony Dispatcher

Standard SMS transmits cleartext 7-bit characters through carrier SMSCs. WeResence bypasses normal SMS handling by sending raw encrypted byte envelopes directly to Port 8901:

```
+-------------------------------------------------------------------------+
| Protocol (1B) | Version (1B) | Key ID (2B) | IV (12B) | Ciphertext (NB) | Tag (16B) |
+-------------------------------------------------------------------------+
```

Native implementation utilizes Android's `SmsManager.sendDataMessage()`:

```
smsManager.sendDataMessage(targetPhone, null, 8901.toShort(), binaryEnvelope, null, null)
```

3. Hardware & OS-Level Defenses

- **Window Shield:** `window.setFlags(FLAG_SECURE, FLAG_SECURE)` enforced at the SurfaceFlinger level. Blocks screenshots and screen recorders.
- **CallLog Sanitizer:** `CallLogManager.scrubCallLogForNumber()` queries `CallLog.Calls.CONTENT_URI` and deletes matching records upon hangup.
- **Anti-Dumping RAM Wipe:** Sensitive key byte arrays are zeroized with `Arrays.fill(bytes, 0)` when the app locks.
- **ADB Protection:** `android:allowBackup="false"` strictly blocks ADB backup extraction.

4. Telecom Edge Cases & Recovery Logic

Scenario	System Behavior
No SIM Recharge, but on Wi-Fi	Auto-detects Wi-Fi as ONLINE. Full VoIP calls and chats operate 100% free with zero carrier balance needed.
No SIM Recharge & No Wi-Fi (Within 60-Day TRAI Window)	Incoming SMS delivered on Port 8901 (TRAI mandates incoming SMS). Outgoing reply held in Smart Outbox until Wi-Fi is reachable.
Offline Voice Call with 0 Balance	Dispatches silent 1-byte SMS ping: "*Missed call ping. Connect to Wi-Fi for free call*".